

Big Mountain Resort: A Pricing & Investment Analysis

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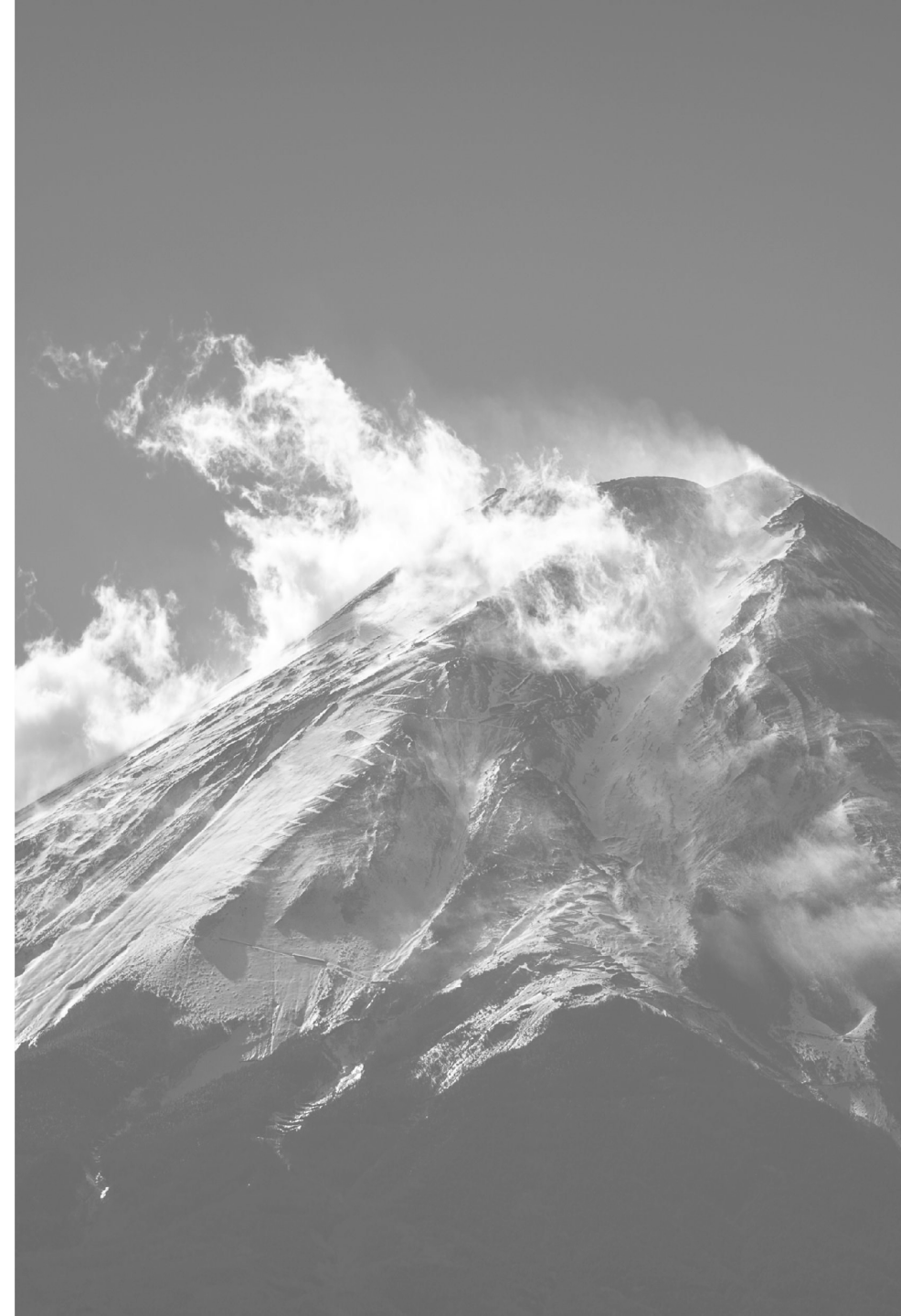


Agenda

1. Problem Identification
2. Recommendation and Key Findings
3. Modeling Results and Analysis
4. Summary and Conclusion

Problem Identification

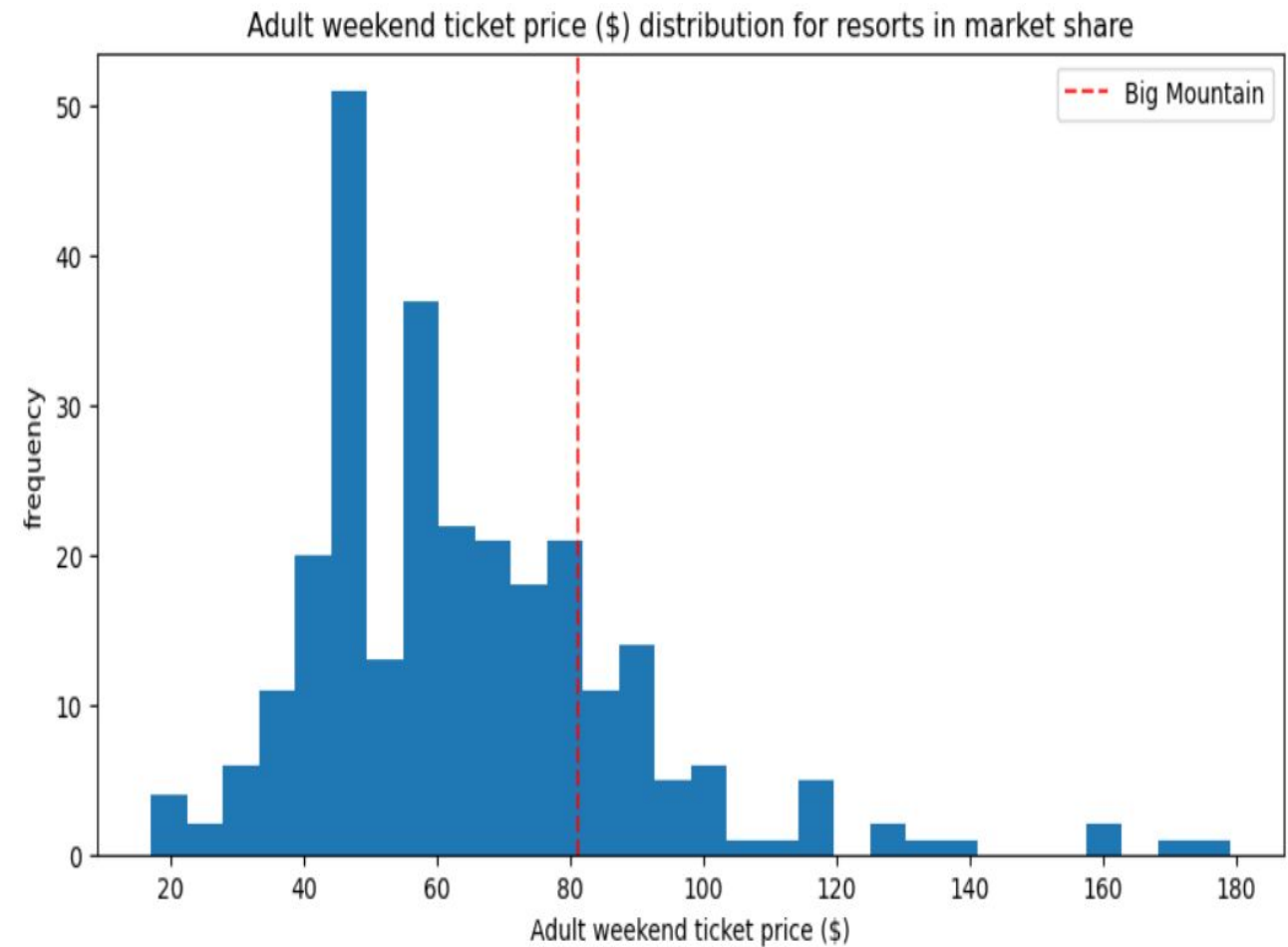
- Big Mountain Resort has traditionally charged premium ticket prices compared with similar ski resorts.
- A new chair lift added **\$1.54 million** in annual operating costs, prompting management to evaluate whether current ticket prices reflect the value of the resort's facilities.
- The goal is to increase revenue while remaining competitive.
- Understanding which resort features drive ticket prices helps management make better pricing and investment decisions.





Goal: Support pricing and investment decisions using data rather than intuition.

Figure 1: Big Mountain Resort is priced below the peer resort distribution, suggesting room for a gradual price increase.



Recommendations and Key findings

1. Gradually increase ticket price

- Gradually increase the adult weekend ticket price toward the model-supported price of \$95.87 while monitoring visitor demand.

2. Continue investing in high-value infrastructure

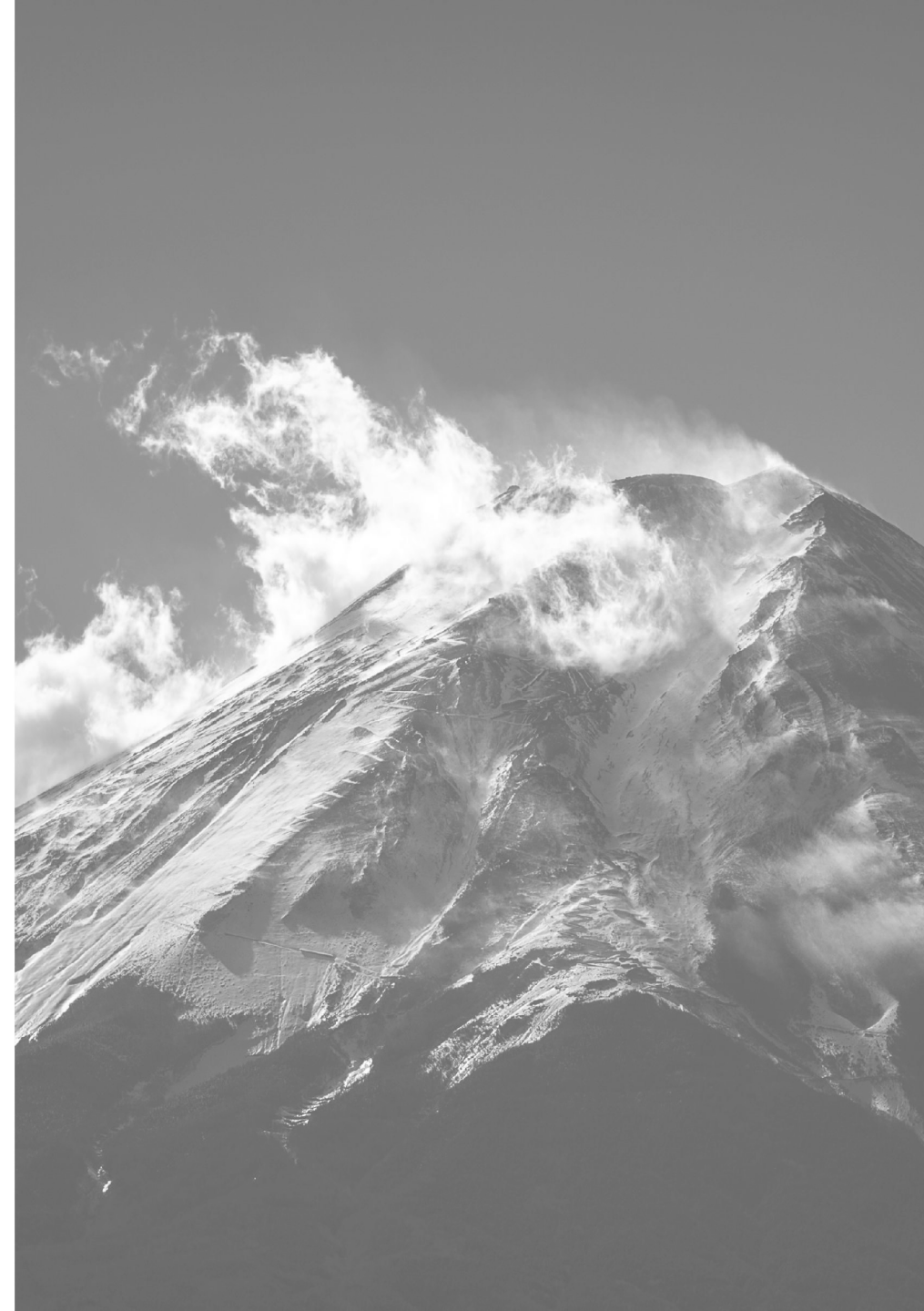
- Scenario 2 (impact of adding one run, increasing the vertical drop by 150 feet, and installing an additional chair lift) generates enough additional revenue to cover the chair lift's annual operating cost.

3. Prioritize Scenario 2

- Best balance of investment and revenue gain.

4. Use run closures cautiously

- If cost reductions are needed, temporarily close low-use runs and monitor visitor demand before making permanent changes.





Modeling Results and Analysis

- **Baseline Benchmark**

A Dummy Regressor that always predicts the average price achieved a test MAE of approximately **\$19**, providing the benchmark that all predictive models must outperform.

- **Linear Regression Results**

The Linear Regression model explained **72%** of ticket-price variation with a test MAE of **\$11.79**. Feature selection reduced overfitting by retaining the eight most informative features.

Key price drivers:

- 1) vertical drop,
- 2) snowmaking acreage,
- 3) total chairs,
- 4) fast quads,
- 5) number of runs, and
- 6) longest run.
- 7) number of trams
- 8) skiable terrain acreage

Cross-validated MAE of **\$10.50 ± \$1.62** closely matched the test error, indicating reliable performance.

Modeling Results and Analysis

- **Random Forest Results**

Random Forest performed best overall: cross-validated MAE of **\$9.64 $\pm$ \$1.35** and a test MAE of **\$9.54**. The close agreement between cross-validation and test results indicates good generalization.

The most influential features were **fast quads**, **number of runs**, **snowmaking acreage**, and **vertical drop**, matching the Linear Regression findings.

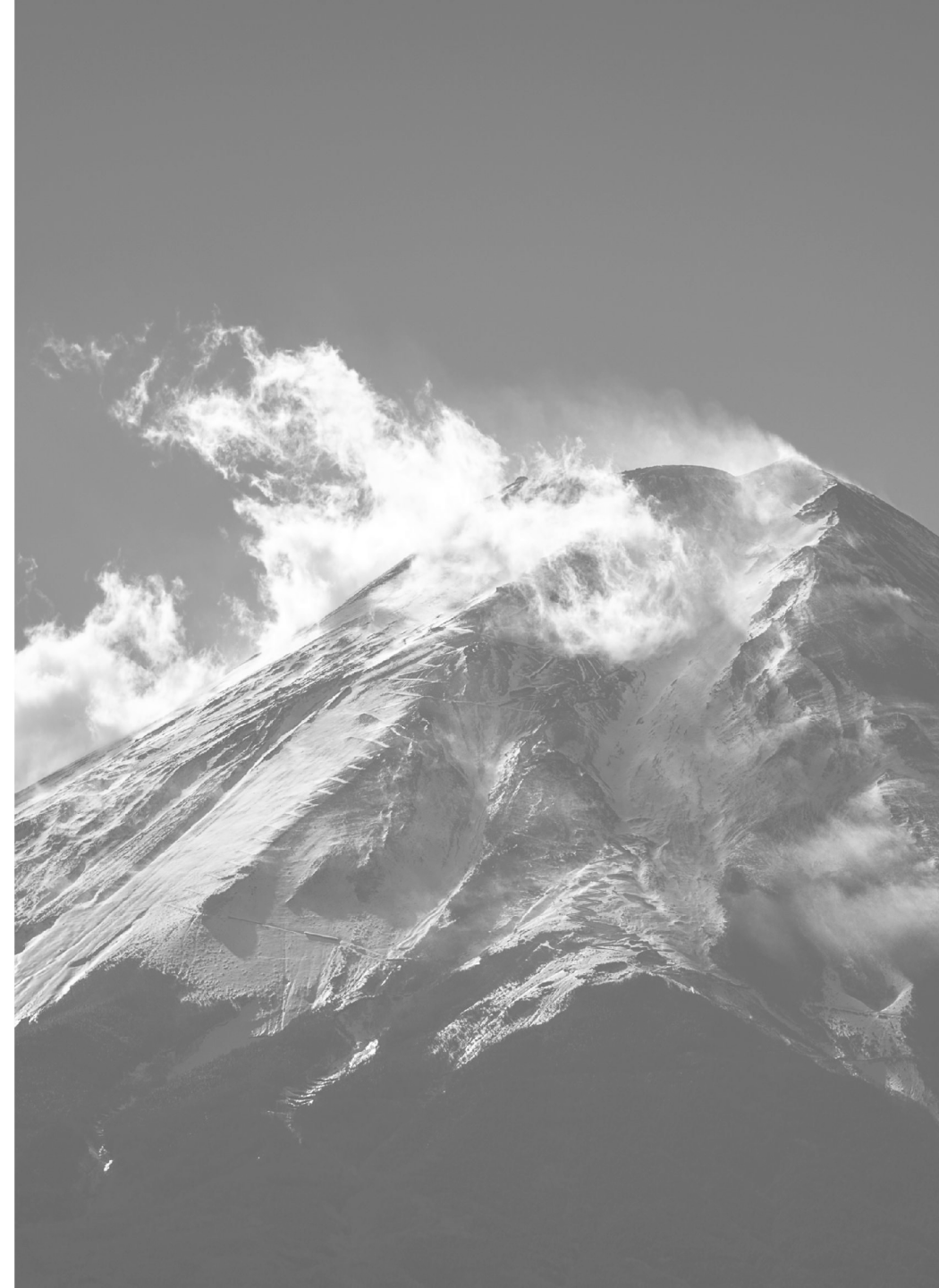
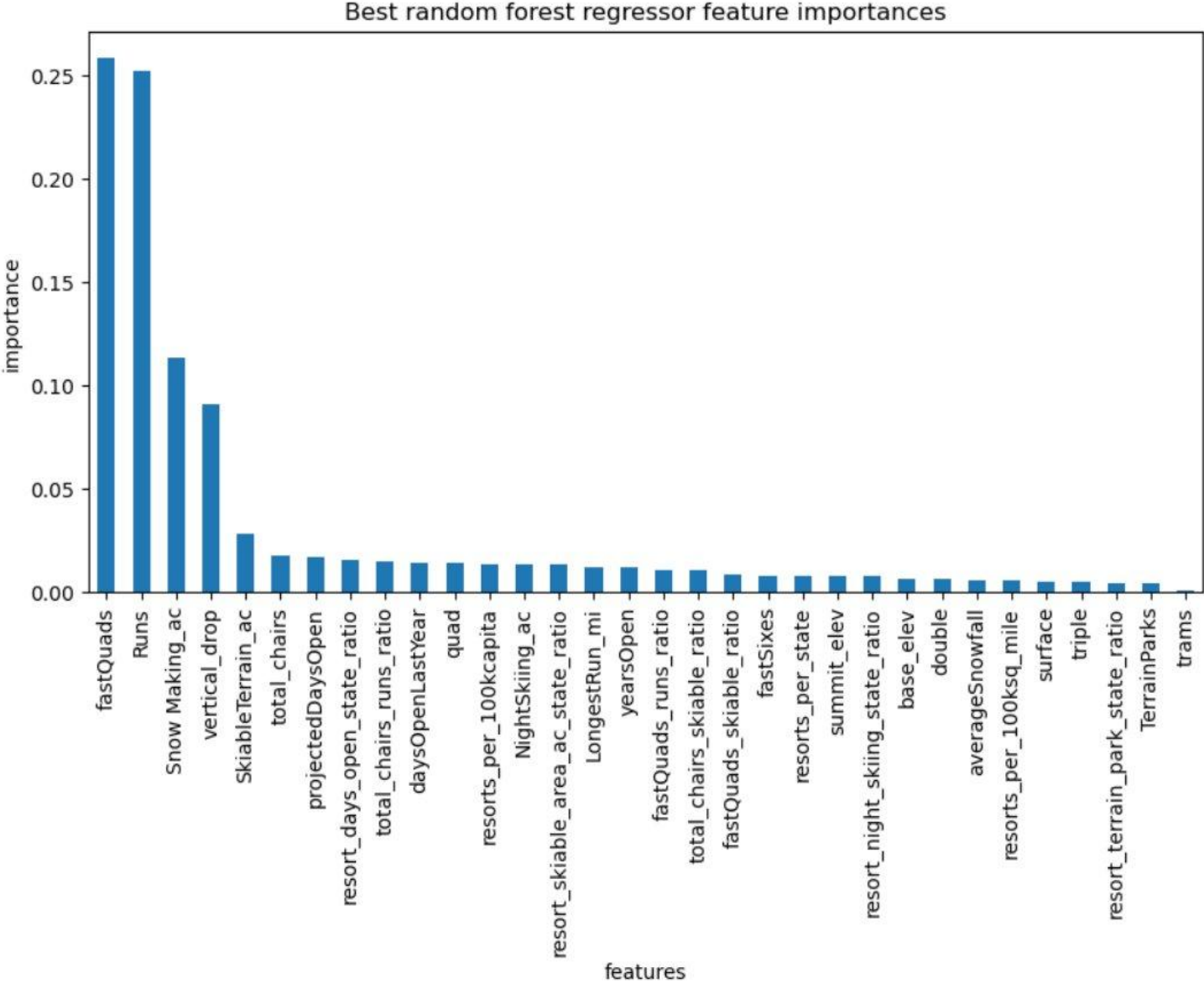




Figure 2: Random Forest Feature Importance



Modeling Results and Analysis

- **Model Performance Comparison**

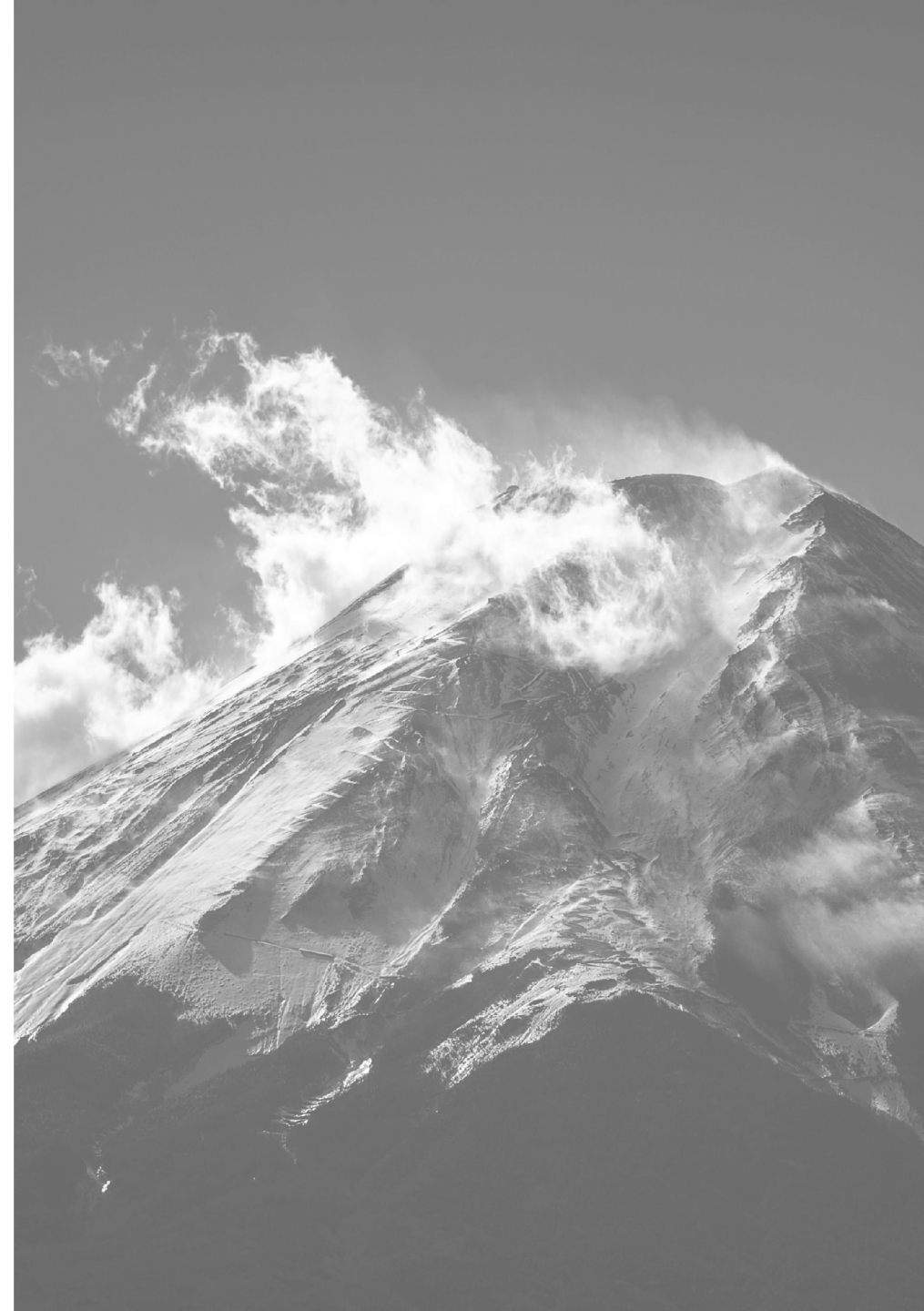
Model performance improves quickly with additional training data before leveling off after about **40–50 observations**, suggesting the current dataset is sufficient.

Figure 3: Learning curve



- **Final Model Selection**

Random Forest was chosen as the final model: lowest error, consistent cross-validation and test performance, and the same key pricing factors as Linear Regression, increasing confidence in the identified pricing drivers.





Summary & Conclusion

Key Findings

- Big Mountain appears to be underpriced relative to comparable resorts.
- A gradual ticket price increase is supported by the model.
- Scenario 2 provides the highest return on investment.

Expected Business Impact

- Higher annual revenue with minimal demand risk.
- More informed pricing and investment decisions.

Limitations & Future Work

- Include operating and capital cost data.
- Update the model with new seasonal data.

Overall, the analysis supports a gradual ticket price increase and continued investment in high-value infrastructure to increase revenue while maintaining competitiveness.